

# Prompt Instruction Versus Code Enforcement for Evidence-Grounded Citation: A Four-Arm Pilot Ablation

## Abstract

This brief is a pilot ablation on evidence grounding, not a study of the original retrieval-augmented-generation question. All empirical claims in it are unverified pending a confirmed experiment-run Evidence entry. We describe a four-arm ablation with eight generations per arm on a single research question, in which the arms vary whether Evidence is present in the stage input and how strongly citation is required, together with a citation instruction, schema enforcement, and a repair retry. The reported pilot signals — coverage of 1.00 and no fabricated ids with Evidence present, roughly tenfold higher declared uncertainty under the citation instruction, minimal schema-enforcement effect with no repair trigger, and, without Evidence, all-fabricated ids, seven-of-eight enforcement rejections, and zero repairs — remain unverified single-question observations. None of the three human-confirmed background papers directly addresses retrieval-augmented-generation provenance.

## 1 Introduction

Retrieval-augmented generation pipelines can emit citations that do not correspond to any retrieved source. Two broad remedies are available. The first is a prompt-level citation instruction: the model is asked to cite its Evidence. The second is code-level enforcement: a schema check rejects generations whose cited identifiers do not resolve, optionally followed by a repair retry. This brief motivates a pilot ablation that contrasts the two mechanisms as ways to suppress fabricated citations. The pilot is deliberately narrow, and the contrast between declared honesty and accepted output is the object of study rather than a completed finding.

## 2 Related Work

Three human-confirmed paper abstracts supply the background for this brief.

The abstract of the causality-benchmark review, `paper-50abf243f9f4e962403b`, states that many existing benchmarks for large language model causal inference and reasoning can likely be solved through retrieval of domain knowledge, which the review says questions whether those benchmarks achieve their intended purpose (`claim.review-retrieval-shortcut`, confidence 0.9). The same abstract reports that recent benchmarks move toward a more thorough definition of causal reasoning by incorporating interventional or counterfactual reasoning and that it derives criteria a useful benchmark should satisfy; the abstract does not enumerate those criteria, so any operational checklist derived from them is an extrapolation (`claim.review-interventional-criteria`, confidence 0.85).

The abstract of the cyber threat attribution paper, `paper-bcc49f7513ed61be3a1f`, proposes a modular architecture that can combine concrete attributors via opinion pools, including a Pairing

Aggregator that sequentially applies logarithmic and linear opinion pools, and reports experimental validation suggesting the modular approach does not decrease performance and can improve precision and recall; this is background by analogy only and does not address retrieval-augmented-generation provenance (`claim.attribution-aggregation-background`, confidence 0.85).

The abstract of the distributed attribute representation paper, `paper-b2b2f01594862b01e12f`, proposes a third-order multiplicative model in which word context and attribute vectors interact to predict the next word, and reports tasks including sentiment classification, cross-lingual document classification, and blog authorship attribution; this is background by analogy only and does not address retrieval-augmented-generation provenance (`claim.attribute-representation-background`, confidence 0.85).

Taken together, none of the three human-confirmed paper abstracts supplied in this project directly addresses retrieval-augmented-generation provenance or claim-to-source traceability, so they can serve only as background and cannot substantiate the empirical findings of this study (`claim.no-rag-provenance-paper`, confidence 0.8).

### 3 Method

The study is a four-arm ablation with eight generations per arm on a single research question. The arms vary whether Evidence is present in the stage input and how strongly citation is required. In the Evidence-present condition the stage input contains the evidence items; in the no-Evidence condition it does not. The citation instruction asks the model to cite Evidence. Schema enforcement checks the generated structure and rejects a generation that fails the schema. A repair retry re-invokes the model after a rejection. The sequence flow of the pipeline under study is shown in Figure 1, and its module graph is shown in Figure 2.

The paired, intervention-style logic of this design is only analogically motivated by the human-confirmed review abstract `paper-50abf243f9f4e962403b`, which reports that stronger benchmarks incorporate interventional or counterfactual reasoning. The actual four-arm design is not recorded in any confirmed Evidence entry, and transposing that rationale from causal-reasoning benchmarks to claim-level provenance is an unvalidated analogy (`claim.ablation-design`, confidence 0.4).

### 4 Results

With Evidence present in the stage input, claim-evidence coverage was 1.00 in every arm and no cited id was fabricated, indicating that adding the citation requirement imposed no measurable citation cost in that condition (`claim.coverage-with-evidence`, confidence 0.25). Instructing the model to cite Evidence raised declared uncertainty roughly tenfold relative to the condition without that instruction (`claim.instruction-raises-declared-uncertainty`, confidence 0.25). Schema enforcement added almost nothing beyond the prompt instruction and never triggered a repair in the Evidence-present arms (`claim.schema-enforcement-minimal`, confidence 0.25). With no Evidence in the stage input, every cited id was fabricated, enforcement rejected seven of eight generations, and the repair retry never repaired one (`claim.no-evidence-fabrication`, confidence 0.25). The no-evidence arm is the only arm in which fabrication was possible by construction, so this rate is not comparable across arms. The repair retry could not repair a model that had nothing to cite, repairing zero of the rejected no-evidence generations (`claim.retry-repaired-none`, confidence 0.25).

Each of these is a single-question pilot observation. The experiment-run Evidence entry that would confirm them is pending/unverified and was therefore not used as claim evidence. Figure 3

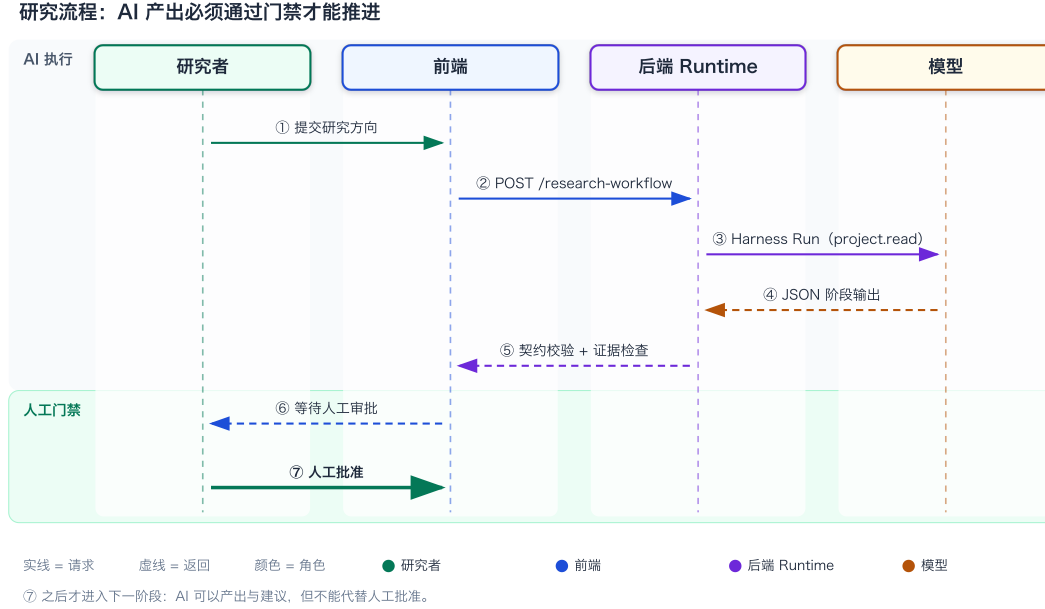


Figure 1: Sequence flow of the pipeline under study: the stage input, the optional Evidence condition, generation, the citation instruction, schema enforcement, and the repair retry.

presents the pilot results.

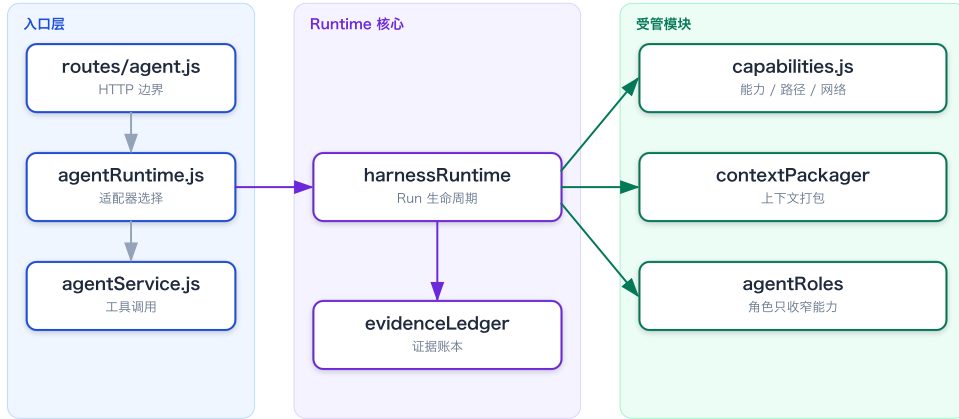
## 5 Analysis

Prompt instruction changes declared honesty, whereas code enforcement is what stops a fabricated citation from being accepted, and the retry cannot repair a model that has nothing to cite (`claim.instruction-vs-enforcement`, confidence 0.2). This interpretation rests on an eight-generations-per-arm, single-question pilot and is not backed by any confirmed Evidence entry, so it must not be reported as a verified result.

## 6 Limitations

- The sample is eight generations per arm on a single research question, so the observed effects are a pilot signal rather than a population estimate; no uncertainty interval, variance summary, or significance test is available.
- Only the no-evidence arm could fabricate citations by construction, so fabrication rates are not comparable across arms and the seven-of-eight rejection figure applies only to that arm.
- The experiment-run Evidence entry `evidence-19e177ec-9a6c-4bac-b897-03c7a6f61b50` is recorded in the stage input ledger with verificationStatus pending and does not resolve as confirmed Evidence in this project, so no empirical claim in this brief cites it or is treated as verified.
- No retrieval-augmented-generation paper is present in the confirmed Evidence set; the three human-confirmed paper entries concern causal-reasoning benchmarks, cyber threat attribution,

Harness 运行路径：一次 Run 经过的模块



只有 harnessRuntime 会调用模型；其余模块只提供约束、上下文与角色。

Figure 2: Module graph of the pipeline under study, showing the components whose interaction the four-arm ablation varies.

and distributed attribute representations, so they are background only and do not directly support evidence-grounding mechanisms.

- Paper 2407.08029v1 is a preprint whose venue, venueLevel, peerReviewed, citationCount, and hasCode fields are unset, and its benchmark criteria are described only at abstract level, so any derived checklist is partly extrapolated.
- The experiment is recorded as planned with a human-approved plan only; no structured Experiment Run has been executed and no unsupported-claim-rate result exists.
- No dataset identity, size, version, license, code artifact, environment specification, or deterministic seed was supplied, so reproducibility cannot be assessed or claimed.
- Ground-truth supported-by-cited-span labels and annotator agreement are unavailable, and the enforcement and repair behavior is observed on a single model and prompt scaffold, so generalization is unestablished.

## 7 Availability and Next Steps

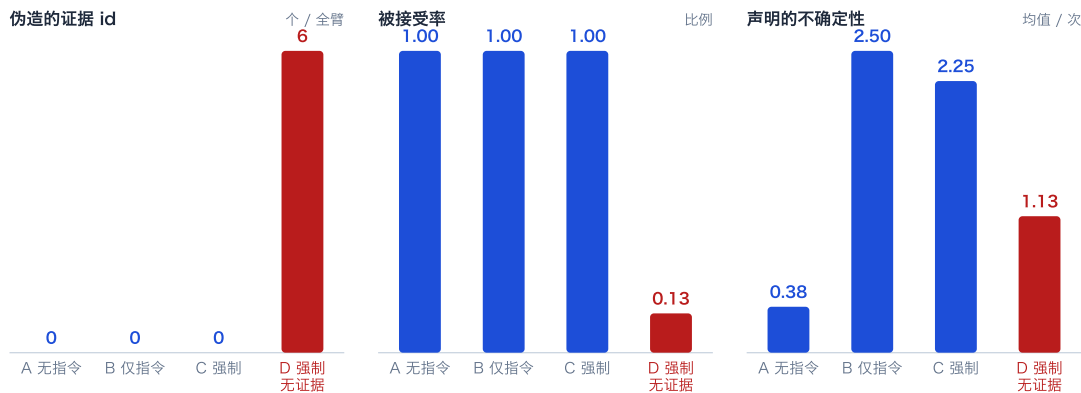
Before any claim in this brief can be promoted to verified, the following human inputs are required: a confirmed experiment-run Evidence entry, dataset and code provenance, and the statistical design. Until those inputs are supplied, every empirical statement above remains a pilot observation.

## 8 Unverified Claims

The following items are unverified and must not be presented as established results.

- **claim.ablation-design**: the four-arm, eight-generations-per-arm design and its interventional logic are only analogically motivated by [paper-50abf243f9f4e962403b](#) and are not recorded in any confirmed Evidence entry.

四臂消融：证据门禁是否改变产出 (n=8 次/臂，模型 global:deepseek-v4.1-flash)



A/B/C 的输入里有 3 篇 human-confirmed 论文可引；D 的输入里没有任何证据。只有 D 存在伪造的可能。

D 臂：6 个引用 id 全部伪造，门禁拒绝 7/8 次，修复重试触发 7 次但修好 0 次。

来源：aidoc/experiment-evidence-gate.json (scripts/experiment-evidence-gate.mjs 产出，严格串行)

Figure 3: Pilot results for the four-arm ablation. The four arms vary whether Evidence is present in the stage input (Evidence present versus no Evidence) and how strongly citation is required (the citation instruction, with schema enforcement applied over it). Bars show the pilot outcome measures reported in the text: claim-evidence coverage, count of fabricated citation ids, declared uncertainty, enforcement rejections, and repairs. All values are single-question, eight-generations-per-arm pilot observations and remain unverified pending a confirmed experiment-run Evidence entry.

- **claim.coverage-with-evidence**: no confirmed experiment-run Evidence entry is present, so the 1.00 coverage and zero-fabrication result is unverified.
- **claim.instruction-raises-declared-uncertainty**: the roughly tenfold rise in declared uncertainty is an unverified single-question pilot observation with no confirmed Evidence entry.
- **claim.schema-enforcement-minimal**: the assertion that schema enforcement added almost nothing and never triggered a repair is unverified with no confirmed Evidence entry.
- **claim.no-evidence-fabrication**: the all-fabricated ids, seven-of-eight rejections, and zero repairs without Evidence are unverified with no confirmed Evidence entry.
- **claim.retry-repaired-none**: the retry repairing zero of the rejected generations is unverified with no confirmed Evidence entry.
- **claim.instruction-vs-enforcement**: the conclusion distinguishing prompt instruction from code enforcement is an unverified interpretation of the pilot and is not a confirmed result.
- The experiment-run Evidence entry `evidence-19e177ec-9a6c-4bac-b897-03c7a6f61b50` is pending/unverified and does not resolve as confirmed Evidence, so it was not cited in any claim and no claim may be reported as verified on its basis.